

Corruption Without Guilt:

An Extended Phenotype Model of Systemic Conflict-of-Interest Failure

Ignacio Adrián Lerer

Independent Researcher | Buenos Aires, Argentina
adrian@lerer.com.ar | ORCID: 0009-0007-6378-9749

Zenodo | July 2026

Keywords: *extended phenotype theory · evolutionary game theory · conflict of interest · anti-corruption · WEIRD psychology · parasitic spontaneous order · heteronomous Bayesian updating · Dennett-Nash gap · memetic replication · institutional design*

ABSTRACT

Anti-corruption frameworks worldwide presuppose that exposure to conflict-of-interest (COI) norms, combined with reputational and legal sanctions, will produce internalized moral constraints in political actors. The empirical record contradicts this assumption with a regularity that demands explanation. Existing scholarship attributes this failure to enforcement deficits or institutional capture. Both explanations treat the symptom and miss the mechanism.

This paper argues that the absence of guilt and shame in political actors who engage in conflict-of-interest behavior is not a pathology but a predictable equilibrium, generated by three mutually reinforcing mechanisms operating at different analytical levels. The first is Parasitic Spontaneous Order (PSO), a macro-systemic process by which favoritism displaces formal norms and generates its own enforceable moral vocabulary in which loyalty is virtue and impersonal compliance is defection. The second is Heteronomous Bayesian Updating (HBU), a meso-level transmission process by which institutional actors form behavioral priors not from outcome evaluation but from the reaction intensity of high-status models; in captured environments, these signals are systematically inverted relative to formal norms. The third is the Dennett-Nash Gap, a micro-level mismatch between anti-corruption tools designed for actors operating at Dennett's Level 3 intentionality (recursive moral processing) and political actors who, in their institutional role, operate at Level 1 (optimization under constraints without recursive moral cognition).

The paper situates this integrated model within the Extended Phenotype Theory of law (EPT) developed in previous work and connects it to Henrich's (2020) WEIRD/non-WEIRD framework to explain why COI norms achieve genuine internalization in a narrow range of cultural environments and produce compliance theater in most others. The analysis generates five falsifiable hypotheses linking the model to existing empirical instruments, including the Constitutional Lock-in Index (CLI) and the Transmission Capacity Index (TCI) developed in prior publications. The central policy implication is that anti-corruption reform designed to produce moral internalization is mis-specified for most institutional environments. Reforms should instead be calibrated to modify the payoff structure at Level 1 while simultaneously building the cultural scaffolding that allows COI norms to achieve memetic fitness over generational timescales.

I. INTRODUCTION: THE GUILT PUZZLE

Picture the scene, repeated across political systems with remarkable fidelity: a senior official sits before a parliamentary committee or a judicial inquiry and, confronted with documented evidence that public decisions consistently benefited private interests, expresses genuine bewilderment at the nature of the accusation. This is not the calculated performance of a sophisticated liar. The perplexity is real. What the inquiry calls a conflict of interest, the official experienced as ordinary loyalty to colleagues, partners, and allies who had, in turn, been loyal in return. The moral grammar of the committee and the moral grammar of the accused are not in tension; they are mutually incomprehensible.

This observation is not new. What is new is the theoretical framework needed to explain it with precision. Anti-corruption scholarship has accumulated decades of evidence that formal COI norms, disclosure requirements, recusal mechanisms, and ethics codes produce dramatically different behavioral outcomes depending on the institutional environment. Transparency International's Corruption Perceptions Index, collected annually since 1995, documents a stable cluster of high-compliance nations concentrated in Scandinavia, New Zealand, Singapore, and the Netherlands, surrounded by a much larger cluster where formal anti-corruption architecture coexists with persistent informal favoritism. The gap between formal norm and actual behavior is not a second-order anomaly waiting to be solved by better enforcement design. It is the primary phenomenon requiring explanation.

The dominant explanatory framework in the corruption literature attributes this gap to two mechanisms: enforcement deficits (sanctions are insufficient to deter rational calculation) and institutional capture (the agencies charged with enforcement are themselves controlled by the actors they are supposed to regulate). Both explanations have empirical support and practical traction. Neither is wrong. But both share a common assumption that this paper contests: that the target actor is, in principle, susceptible to the guilt and shame that the formal system is designed to produce, and that only external failure prevents internalization.

That assumption is incorrect, and its incorrectness is not peripheral but structural. The absence of guilt and shame in conflict-of-interest behavior is not a contingent failure of socialization that better enforcement could remedy. It is the predictable output of a psychological architecture that evolved for a different task, transmitting norms calibrated to a different environment, in interaction with institutional actors whose intentionality structure is systematically misaligned with the tools designed to regulate them. This paper provides the mechanism.

Three mechanisms, operating at distinct analytical levels, jointly produce the equilibrium. At the macro-systemic level, Parasitic Spontaneous Order converts favoritism from an individual strategy into an emergent normative system with its own moral vocabulary, its own sanctions, and its own definition of what constitutes ethical conduct. At the meso-institutional level, Heteronomous Bayesian Updating transmits inverted behavioral priors across generations of political actors, so that each new entrant to the system forms expectations calibrated to what high-status models do rather than what formal norms prescribe. At the micro-interactional level, the Dennett-Nash Gap ensures that anti-corruption tools designed for morally recursive actors fail to register as intended incentives for actors who process institutional signals as pure cost-benefit calculations.

The framework draws on the Extended Phenotype Theory of law developed in Lerer (2024, 2025), which treats legal institutions as extended phenotypes of competing cultural replicators (memes) subject to evolutionary selection. It integrates Henrich's (2020) documented distinction between WEIRD (Western, Educated, Industrialized, Rich, Democratic) and non-WEIRD psychological profiles to explain why COI norms achieve genuine internalization only in a narrow set of cultural environments. And it connects to formal instruments developed in prior work, specifically the Constitutional Lock-in Index (CLI) and the Transmission Capacity Index (TCI), to generate

testable predictions about when the equilibrium described here will hold and when it can be disrupted.

The structure is as follows. Section II establishes the evolutionary substrate of guilt and shame as bounded ingroup technologies, not general-purpose moral sensors. Section III analyzes COI as a universalist meme that requires specific cultural preconditions unavailable in most political environments. Section IV presents the three mechanisms of internalization failure in detail. Section V integrates them into a unified EPT model of corruption as parasitic extended phenotype. Section VI presents the game-theoretic characterization of the stable defection equilibrium in verbal-semiformal terms. Section VII generates five falsifiable hypotheses. Section VIII draws implications for institutional design.

II. THE EVOLUTIONARY SUBSTRATE OF MORAL EMOTIONS

Guilt and shame have a history. That history is not primarily philosophical or cultural; it is biological. Both emotions are adaptations, products of selection pressures operating over hundreds of thousands of years in the specific social environments that shaped human psychology. Understanding what they evolved to do is the necessary first step toward understanding why they do not do what anti-corruption frameworks expect of them.

Robin Dunbar's research on primate group sizes and neocortical volume converged on a number that has become one of the most-cited in social science: approximately 150 individuals, the cognitive limit of stable social relationships that humans can maintain simultaneously (Dunbar 1992). This is not merely a neurological fact about memory capacity. It describes the social world in which the psychological architecture of guilt and shame evolved. For most of human evolutionary history, the morally relevant universe was a band of around 50 to 150 known individuals, with whom one interacted repeatedly, whose reputations were tracked in detail, and whose reciprocal cooperation was the primary determinant of survival and reproduction.

In that world, guilt and shame were functional technologies with precise targets. Shame is a response to actual or anticipated negative evaluation by members of one's social group; it evolved to maintain reputation within the ingroup (Fessler 2007). Guilt is the internal signal that one has violated a cooperative norm with a known partner; it evolved to repair relationships and signal

commitment to future cooperation (Trivers 1971). Neither emotion is designed to operate toward anonymous strangers, abstract institutional roles, or diffuse publics. Their scope is bounded by the cognitive radius of social embeddedness.

William D. Hamilton's theory of inclusive fitness, formalized in a landmark 1964 paper in the *Journal of Theoretical Biology*, provides the genetic logic underlying this bounded scope. Hamilton demonstrated that natural selection favors behaviors that maximize the propagation of shared genes, weighted by the coefficient of genetic relatedness. The mathematical result, known as Hamilton's rule, predicts that organisms will preferentially cooperate with genetic relatives and, by extension, with close allies whose fitness is correlated with one's own. In the ancestral environment, the ingroup was substantially composed of kin or long-term reciprocal partners. Favoritism toward this group was not a violation of fitness logic; it was an expression of it.

The crucial inference is this: the political actor who assigns public resources to allies is not malfunctioning morally. The psychological hardware is operating exactly as evolution designed it to operate. Favoritism toward known partners maximized fitness in the environment of evolutionary adaptedness. The conflict-of-interest concept requires overriding this architecture, substituting procedural neutrality toward strangers for preferential treatment of allies. That substitution is possible; humans are capable of it. But it is a demanding cultural achievement, not a factory setting.

Robert Trivers' (1971) reciprocal altruism model and Robert Axelrod's (1984) computational tournaments on iterated Prisoner's Dilemma confirmed that the psychological emotions supporting cooperation, including guilt over defection and shame at having been caught defecting, are calibrated to repeated-interaction settings with identifiable partners. Axelrod's tournaments demonstrated that the strategy Tit for Tat, which cooperates on the first move and then mirrors the partner's previous action, dominates all competing strategies in iterated games, not because it is morally superior but because it produces the highest payoffs when players interact repeatedly. The emotions that enforce this strategy, guilt about defection, shame about being seen to defect, are precisely calibrated to this iterated known-partner context. They do not generalize automatically to one-shot interactions with strangers, which is the normative structure assumed by COI frameworks.

Michael Tomasello's developmental research at the Max Planck Institute for Evolutionary Anthropology adds a critical developmental dimension. His experiments, replicated extensively across cultural settings, demonstrate that human infants cooperate spontaneously before they can articulate reasons for doing so (Tomasello 2009). Cooperation is the prior; defection requires cognitive development. But cooperation, as Tomasello's data make clear, is directed: infants cooperate with known and trusted agents. The capacity to extend cooperation to anonymous institutional roles, to a tax authority, a public procurement office, or an abstract public interest, is a cognitive and cultural achievement that develops later and requires specific institutional scaffolding.

Three conclusions follow from this evolutionary account, each of them directly relevant to the failure of anti-corruption frameworks. First, the absence of guilt in conflict-of-interest behavior does not indicate that the actor lacks moral emotions; it indicates that the actor's moral emotions are calibrated to a different domain. The official who feels no guilt about awarding a contract to an ally may feel intense guilt about betraying that ally. The emotions are present and functional. They are directed at the wrong target from the perspective of the formal norm. Second, moral emotions are not scalable by mandate. Demanding that political actors extend guilt and shame to anonymous publics is demanding an evolutionary leap that cannot be produced by ethics training or disclosure requirements. Third, the social environments in which formal COI norms can achieve genuine internalization, those in which procedural neutrality toward strangers is the default psychological orientation, are evolutionarily recent and culturally specific. They require explanation, not assumption.

III. CONFLICT OF INTEREST AS A UNIVERSALIST MEME

The concept of conflict of interest, as a freestanding legal and ethical category, is not ancient. The idea that an individual's private interests might systematically distort their exercise of public authority has ancient precursors, traceable to Aristotle's analysis of civic corruption and Roman republican prohibitions on certain commercial activities by magistrates. But the formalization of COI as a distinct normative concept, subject to its own compliance architecture, its own disclosure regimes, and its own administrative enforcement mechanisms, is a product of

nineteenth and twentieth century liberal institutional design. It crystallized in the United States in the post-Civil War period, spread to Western Europe through administrative law reform in the early twentieth century, and was transmitted globally through OECD conventions, World Bank conditionality, and post-Cold War governance reforms in the 1990s.

As a meme, COI has specific requirements. It does not replicate successfully in all cognitive and institutional environments. Three preconditions, each of them cognitively demanding and culturally specific, must be in place for COI norms to achieve genuine internalization rather than surface compliance.

The first precondition is role-person separation: the capacity to distinguish between the individual and the institutional role she occupies, such that the role carries obligations that override personal relationships and loyalties. This is not a natural psychological category. The evolutionary default is to treat authority as personal, not positional. The tribal chief's authority is his, not the office's; his obligations run to his kin and allies, not to an abstract constituency. Role-person separation requires the internalization of the idea that when one sits in a particular chair, one is temporarily constituted as a different agent with different obligations. This is a sophisticated cognitive achievement, and it requires extensive institutional reinforcement to become habitual.

The second precondition is procedural neutrality toward strangers: the disposition to apply the same rules to known allies and anonymous strangers without differentiation based on the relationship. This directly contradicts the evolved preferential logic described in Section II. It requires not merely tolerance of strangers but active suppression of the differential treatment that ingroup psychology generates. The societies in which this disposition is most deeply embedded, the Scandinavian countries, New Zealand, the Netherlands, share historical features that Joseph Henrich's work identifies as constitutive of the WEIRD psychological profile: centuries of weakened kinship networks, urbanization that created repeated interactions with strangers, and Protestant institutional culture that emphasized impersonal rule-following.

The third precondition is internalized compliance without observation: the capacity to follow COI norms even when no one is watching, because the norm has become constitutive of one's professional identity rather than merely an external constraint. This is the deepest and most demanding of the three preconditions. It requires what Philip Pettit (1997) calls the disposition to track reasons rather than merely respond to incentives. Political actors who comply only when

observed, and calculate whether non-compliance will be detected when unobserved, are not internalizing the COI norm; they are negotiating with it.

Joseph Henrich's *The WEIRD People in the World* (2020) provides the historical account of how these preconditions were produced in a specific subset of human societies. The proximate mechanism was the Catholic Church's kin avoidance program, a set of marriage prohibitions that began to be enforced aggressively in the fifth century and expanded through successive ecclesiastical councils, culminating in prohibitions that extended to sixth-degree relatives by the High Medieval period. This program systematically dismantled the intensive kinship networks that had organized social cooperation across most of Europe and, by extension, forced European populations to develop new mechanisms for cooperating with non-kin: impersonal markets, voluntary associations, professional guilds, and eventually formal institutions with rule-bound procedures.

The psychological consequences, documented by Henrich through a vast program of cross-cultural experiments, were measurable. WEIRD populations show higher individualism, higher impersonal trust, greater willingness to follow rules with anonymous others, and greater comfort with universalist moral reasoning that treats strangers and allies by the same standards. These are not fixed genetic traits; they are psychological profiles shaped by a millennium of specific institutional history. They are also, for the purpose of this analysis, the preconditions for genuine COI internalization.

Most of humanity did not undergo this process. Latin America, sub-Saharan Africa, East Asia, South Asia, and the Middle East maintained intensive kinship networks through and beyond the colonial period, and in many cases colonial administration reinforced rather than dismantled them. The formal institutions transplanted to these regions through colonialism and post-colonial governance reform carried WEIRD assumptions about the cognitive and cultural preconditions for compliance. When those assumptions were false, the institutions produced a characteristic outcome: formal compliance theater, in which disclosure forms are filed, ethics codes are printed and distributed, recusal procedures are nominally available, and actual favoritism continues undisturbed.

The implementation gap data are consistent with this account. Societies classified as WEIRD by Henrich's criteria exhibit gaps between formal legal norms and actual institutional behavior in the

range of 5 to 6 percent, as documented in comparative work on legal transplants (Lerer 2025, citing implementation gap data from World Justice Project Rule of Law Index). Non-WEIRD societies exhibit gaps in the range of 28 to 35 percent. This difference of roughly 25 percentage points is not explained by differences in formal institutional design, which has converged substantially through international norm transmission. It is explained by the presence or absence of the psychological substrate that allows formal norms to achieve behavioral purchase.

The COI meme, transplanted without its underlying cultural infrastructure, does not disappear; it mutates into compliance theater. The disclosure form becomes a ritual that signals awareness of the category without producing behavioral constraints. The ethics officer becomes a bureaucratic actor whose function is to document compliance rather than produce it. The recusal procedure becomes a resource deployed strategically when it is politically convenient and ignored when it is not. These are not random failures; they are the predictable adaptive responses of a meme that has landed in an environment where it cannot achieve the fitness its design requires.

IV. THREE MECHANISMS OF INTERNALIZATION FAILURE

The evolutionary substrate and the cultural transplant failure analyzed in the preceding sections provide the background conditions. This section identifies the three proximate mechanisms through which these background conditions produce the specific equilibrium of guilt-free conflict-of-interest behavior. The mechanisms operate at different analytical levels and reinforce one another, but each has a distinct logic that can be specified independently.

IV.A. Parasitic Spontaneous Order

Friedrich Hayek's concept of spontaneous order describes the emergence of functional social arrangements that no individual designed and no authority decreed (Hayek 1973). The price system in competitive markets is the paradigm case: no central coordinator produces the information-aggregating function of market prices; it emerges from the decentralized actions of millions of agents pursuing their own ends. Hayek's insight was that many of the most functional social institutions, including common law, language, and the norms of commercial practice, are of this emergent character.

Hayek's framework is descriptively neutral: it characterizes emergence, not fitness. An order can emerge spontaneously and be extractive rather than productive. When the dominant strategy in a political system is favoritism, and when formal enforcement is too weak to make defection from favoritism systematically costly, the interactions of many actors each pursuing the loyalty strategy generate an emergent normative order with its own structure: the Parasitic Spontaneous Order. The EPT framework developed in Lerer (2024) designates this as a subcase of spontaneous order in which the emergent arrangements extract resources from the formal system without producing systemic value, and in which the meme driving the order is loyalty rather than impersonal cooperation.

The Parasitic Spontaneous Order has three constitutive features that distinguish it from mere individual favoritism. First, it generates its own enforcement mechanisms. Loyalty is not merely advantaged; disloyalty is actively sanctioned. The political actor who recuses from a decision benefiting an ally, or who reports a colleague's COI violation to an oversight body, is not merely failing to exploit an opportunity; she is defecting from a cooperative arrangement and can expect reputational and material consequences within the ingroup. The enforcement of the PSO's norms is often more reliable than the enforcement of formal anti-corruption norms.

Second, the PSO generates its own moral vocabulary. Within the PSO's normative frame, loyalty is not a failure of institutional integrity; it is the expression of the deepest social virtue. Betraying a colleague to an oversight body, even when that colleague has clearly violated a formal rule, is not compliance with an ethical obligation; it is treachery. The actor who reports the violation has violated a more fundamental norm. This vocabulary is not hypocritical; it is the natural output of the evolved ingroup psychology described in Section II, operating through the transmission mechanism described in Section IV.B below.

Third, the PSO achieves stability through a self-reinforcing dynamics. Each actor who plays the loyalty strategy raises the cost of playing the COI-compliance strategy for all other actors, because COI-compliance without reciprocity from others produces inferior outcomes. When the PSO reaches sufficient density, COI compliance becomes not merely disadvantageous but socially costly: it marks the compliant actor as untrustworthy within the ingroup, as someone who prioritizes abstract rules over concrete relationships. The PSO is in this sense autocatalytic; once

it crosses a critical threshold of density, it becomes self-sustaining without requiring coordination or central direction.

The three activation conditions for PSO identified in the EPT taxonomy (Lerer 2024, *tabla maestra* EPT) are: visible inequality in the distribution of public resources; organized intermediaries who coordinate loyalty networks; and weakened formal enforcement. When all three conditions are simultaneously present, the PSO achieves the stability threshold beyond which unilateral deviation to COI compliance is individually irrational. This is not a contingent political problem; it is a game-theoretic equilibrium, which is to say it is a configuration from which no individual actor has an incentive to deviate.

IV.B. Heteronomous Bayesian Updating

Political actors do not form behavioral norms through direct outcome evaluation. The environments in which they learn are too complex, the feedback loops too delayed, and the causal chains too opaque for direct learning to be effective. They learn primarily by observing what high-status actors do and calibrating their expectations accordingly. This is heteronomous Bayesian updating: prior formation through reaction intensity observed from authority figures, rather than through personal outcome evaluation (Lerer 2025a).

The mechanism operates as follows. A new entrant to a political institution observes the behavior of senior colleagues who serve as models. She does not merely observe what behaviors produce what outcomes; she observes the intensity of social reactions to those behaviors. A formal COI violation that produces a mild reprimand, or no reaction at all, carries a very different signal than one that produces immediate and severe sanction. The prior formed is calibrated to the intensity of the reaction, not the formal category of the behavior.

In institutional environments where COI violations by high-status actors are routinely tolerated, the prior formed by new entrants within one to two years of institutional socialization is that COI violations are acceptable provided they are managed with appropriate discretion. This is not a cynical calculation; it is Bayesian rationality applied to observed social signals. The actor who internalizes 'COI is tolerated here' is doing exactly what the cognitive architecture is designed to do: updating beliefs on the basis of observed evidence.

The perverse consequence is that HBU operates with equal efficiency in either direction. In institutional environments where COI violations produce high-intensity negative reactions from respected authorities, the prior formed is that COI violations are genuinely prohibited and actually enforced. The same cognitive mechanism that transmits tolerance in captured institutions transmits commitment to institutional integrity in well-functioning ones. The mechanism is not intrinsically corrupt; it is intrinsically sensitive to the signals it receives.

What distinguishes captured institutions is not that HBU operates differently; it is that the signals it receives are inverted relative to formal norms. The formal rule says: COI violations are prohibited and will be sanctioned. The observed signal says: COI violations by senior actors are tolerated and sometimes rewarded. When these two inputs conflict, HBU resolves the conflict in favor of the observed signal, because observed behavior by high-status actors is, from a Bayesian perspective, vastly more reliable evidence about the actual normative structure of the institution than formal text.

The four-stage HBU transmission process documented in prior work on constitutional lock-in is directly applicable here. Stage 1: new entrants observe the behavior of authority figures. Stage 2: reaction intensity from those figures forms strong behavioral priors. Stage 3: those priors generate extended phenotypes, the behavioral patterns and informal institutional arrangements that express them. Stage 4: when formal norms contradict the established priors, active inference dominates perceptual inference; the actor acts to make the environment conform to the prior rather than updating the prior to conform to the environment (Lerer 2025a, citing Friston 2010 on the free energy principle). The result is the institutional calcification of COI tolerance, reproduced generationally without requiring any actor to consciously choose to transmit corrupt norms.

IV.C. The Dennett-Nash Gap

Daniel Dennett's (1987) taxonomy of intentional levels provides the analytical vocabulary for the third mechanism. Level 1 intentionality describes agents who optimize against fixed constraints without modeling the mental states of other agents; a thermostat or a simple predator exemplifies Level 1. Level 2 describes agents who model the beliefs and desires of others but do so without recursive embedding: she thinks he wants X. Level 3 describes agents with full recursive intentionality: she thinks he thinks she wants Y, and she anticipates how he will reason

about her reasoning. Human adults operating in normal social contexts are Level 3 agents. The capacity for shame, which requires modeling the imagined evaluations of others who are themselves modeling one's own conduct, is intrinsically Level 3.

Anti-corruption frameworks are designed by Level 3 agents for Level 3 agents. Ethics codes assume that the target actor will read the text, model the intentions of its authors, and internalize the norm because she understands that the norm is part of a cooperative arrangement that benefits all parties, including herself, over time. Disclosure requirements assume that the target actor will anticipate the shame of public exposure and be deterred by that anticipation. Reputational sanctions assume that the actor processes social reproof recursively and is affected by what others think about her.

The problem is not that political actors in high-corruption environments are intrinsically Level 1. They are Level 3 agents in their personal and ingroup contexts: they experience shame, they model others' mental states, they reason recursively about social relationships. The mismatch is role-specific. When operating within the PSO described in Section IV.A, and after having formed the inverted priors described in Section IV.B, political actors process formal institutional signals, including ethics codes, disclosure requirements, and oversight mechanisms, as Level 1 inputs: cost-benefit calculations about the probability and severity of formal sanction. The recursive moral processing that would generate guilt and shame is directed toward the ingroup, not toward the abstract public interest served by the formal institution.

This is the Dennett-Nash Gap as formalized in Lerer (2025b): the payoff differential that an actor operating at actual intentionality Level 1 in an institutional interaction obtains relative to the Nash equilibrium expected under homogeneous Level 3 assumptions. The formal expression is $\Delta_i = SA_i + ICS_i - MP_i + SIP_i$, where SA is the strategic advantage from being immune to Level 3 sanctions (shame, moral reproof), ICS is the ingroup cooperation surplus gained from the loyalty strategy, MP is the moral processing cost avoided by operating at Level 1 in the formal domain, and SIP is the social immunity premium conferred by the PSO (Lerer 2025b).

Three implications follow. First, anti-corruption tools that operate exclusively through the Level 3 channel (shame, moral suasion, reputational effects) will produce no behavioral change in actors who have redirected their Level 3 processing toward ingroup rather than institutional obligations. Second, the gap is not fixed: it narrows when formal sanctions become sufficiently severe to

register as Level 1 costs, when the ingroup itself internalizes COI norms (shifting the loyalty calculation), or when common knowledge cascades, in Steven Pinker's (2025) sense of knowledge that everyone knows that everyone knows, make COI compliance the dominant strategy regardless of intentionality level. Third, the gap produces a systematic asymmetry: the Level 3 actor loses to the Level 1 actor in interactions where formal enforcement is unreliable, because the Level 3 actor bears the cost of moral processing (guilt about potential defection, restraint imposed by anticipated shame) that the Level 1 actor does not pay.

V. THE EXTENDED PHENOTYPE THEORY (EPT) MODEL: CORRUPTION AS PARASITIC EXTENDED PHENOTYPE

The three mechanisms described in Section IV do not operate as independent variables. They constitute a unified system in which each mechanism creates the conditions for the others to operate at full efficiency. The PSO generates the inverted normative environment that makes HBU transmit tolerance rather than integrity; HBU transmits the priors that sustain the PSO across generations without requiring active coordination; and the Dennett-Nash Gap ensures that formal interventions designed to disrupt the PSO and correct the HBU-transmitted priors fail to achieve their intended effects because they are calibrated to the wrong intentionality level. Understanding the system requires a framework that can accommodate all three levels simultaneously. The Extended Phenotype Theory of law provides that framework.

Richard Dawkins' core insight in *The Extended Phenotype* (1982) was that the unit of selection is the replicator, not the organism. Genes are replicators; organisms are their vehicles. What makes genes the relevant unit of analysis is that they persist across generations, replicating with variation, while organisms are temporary structures that serve the replication process. The extended phenotype extends this insight: genes do not merely build bodies; they build artifacts in the external environment, beaver dams, bird nests, spider webs, that enhance the replicator's fitness. The dam is as much a product of genetic selection as the beaver's tail.

Daniel Dennett extended this framework to cultural evolution (Dennett 1995). Memes, units of cultural information that replicate through imitation and transmission, are the cultural analogue of genes. They build extended phenotypes: institutions, practices, rituals, and normative systems that

enhance their own replication. A legal doctrine that generates a body of supportive case law, a tenured faculty, a network of practicing specialists, and a volume of academic commentary has constructed an institutional extended phenotype that dramatically enhances its own survival. The doctrine replicates not because it is true or just but because it has built structures that make its replication efficient.

Applied to the corruption equilibrium described in this paper, the EPT model identifies the loyalty meme as the replicating unit and the PSO as its extended phenotype. The loyalty meme, the cultural complex of attitudes, expectations, and norms that makes favoritism the default strategy and impersonal compliance the deviation, replicates through HBU: each generation of political actors is socialized into it by observing the behavior of predecessors. It builds extended phenotypes in the form of the informal institutional arrangements that constitute the PSO: captured oversight bodies, loyalty-based appointment processes, informal enforcement networks, and the moral vocabulary that makes loyalty a virtue. These extended phenotypes, in turn, make the loyalty meme more fit in the next generation, because they shape the signals that new entrants observe.

The competing meme is the COI norm: the cultural complex of attitudes, expectations, and norms that makes procedural neutrality the default and favoritism the deviation. As Section III argued, this meme requires specific cultural preconditions, the WEIRD psychological substrate, to achieve fitness. In environments where those preconditions are absent, the COI meme arrives without the institutional scaffolding needed to build competitive extended phenotypes. It produces formal artifacts, ethics codes, disclosure requirements, oversight offices, but these artifacts do not generate the institutional infrastructure that would make COI compliance the dominant strategy. They are memes without functioning extended phenotypes; the artifacts exist, but they do not replicate the meme effectively.

The fitness differential between the loyalty meme and the COI meme in non-WEIRD environments is not primarily a function of the intrinsic quality of either meme's design. It is a function of the extended phenotype structures each has managed to build. The loyalty meme's extended phenotypes, the PSO and the HBU transmission chain, are well-adapted to the environment because they evolved in it. The COI meme's extended phenotypes were designed for a different environment and transplanted without adaptation. This is a legal transplant failure, but it operates

at the deepest possible level: not the transplant of a formal rule into a different legal system, but the transplant of a meme into an environment where its fitness requirements are not met.

The EPT model generates a specific prediction about what reform would need to achieve to shift the equilibrium. Reform cannot work by improving the formal artifacts of the COI meme, producing better ethics codes, more detailed disclosure requirements, more numerous oversight offices, as long as the loyalty meme's extended phenotypes remain intact. Reform must target the extended phenotypes of the loyalty meme directly: the PSO's density (by reducing the ratio of private to public benefit from loyalty strategies), the HBU transmission chain (by changing the signals that high-status models emit), and the Dennett-Nash Gap (by raising formal sanction costs to the Level 1 threshold at which they register as material constraints regardless of intentionality level). This is not a modest reform agenda. But it is the correct diagnosis of what is required, and it is better than the current trajectory of prescribing stronger doses of a medicine that does not treat the actual disease.

VI. EVOLUTIONARY GAME THEORY (EGT) FORMALIZATION: THE STABLE DEFECTION EQUILIBRIUM

The preceding sections have established the mechanisms through which guilt-free COI behavior is produced and sustained. This section characterizes the resulting equilibrium in evolutionary game theory terms, using verbal-semiformal analysis to make the structure transparent without requiring full mathematical treatment. For a complete formal exposition of the underlying framework, including explicit payoff matrices and replicator dynamics, the reader is directed to Lerer (2025b), which develops the Dennett-Nash Gap formalism, and to Lerer (2025c), which applies analogous payoff analysis to the constitutional reform equilibrium.

Consider two strategies available to political actors in a high-PSO environment. The first strategy, COI-Compliance, involves following formal conflict-of-interest rules: recusing from decisions that benefit private interests, disclosing relevant relationships, refraining from loyalty-based resource allocation. The second strategy, Loyalty-Optimization, involves prioritizing the allocation of benefits to ingroup allies, accepting returns from those allies, and managing formal COI requirements minimally.

The payoff structure in a high-PSO, low-enforcement environment has a characteristic shape. A Loyalty-Optimizer facing another Loyalty-Optimizer receives the full ingroup cooperation surplus: access to alliance networks, reciprocal resource flows, and the social protections that the PSO affords its members. A COI-Compliant actor facing a Loyalty-Optimizer receives no ingroup cooperation surplus (having been identified as an unreliable partner), bears the cost of moral processing (the psychic and practical effort of resisting loyalty pressures), and receives no formal protection from the system (which enforcement is too weak to provide). A COI-Compliant actor facing another COI-Compliant actor receives a coordination payoff: mutual adherence to formal rules produces predictable interactions and some degree of institutional protection. A Loyalty-Optimizer facing a COI-Compliant actor captures the full strategic advantage: the Loyalty-Optimizer receives ingroup benefits while the COI-Compliant actor provides no competitive threat.

The outcome is a dominant strategy equilibrium for Loyalty-Optimization. Unlike the classic Prisoner's Dilemma, where both mutual cooperation and mutual defection are possible equilibria depending on discount rates and iteration, the PSO transforms the payoff structure so that Loyalty-Optimization dominates COI-Compliance regardless of what other actors do. When PSO density is high, the COI-Compliant actor is worse off in every interaction: worse against Loyalty-Optimizers (no ingroup benefits, no formal protection), and worse against other COI-Compliant actors (mutual compliance generates lower returns than the ingroup cooperation that Loyalty-Optimization affords).

This is an Evolutionarily Stable Strategy in Maynard Smith's (1982) sense: a strategy that, when adopted by all members of a population, cannot be invaded by a mutant strategy. A population of Loyalty-Optimizers cannot be successfully invaded by COI-Compliant mutants, because the mutants are worse off in every interaction with the incumbents. The equilibrium is stable not because it is efficient, it is not, but because no individual actor has an incentive to deviate from it unilaterally.

The conditions for equilibrium collapse are worth specifying, because they identify the points at which reform can achieve traction. The equilibrium breaks down when any of three threshold conditions is met. The first is the sanction cost threshold: when formal sanctions become severe enough to register as Level 1 material costs, even actors who are directing their Level 3 processing

toward the ingroup will calculate that Loyalty-Optimization is more costly than COI-Compliance in expectational terms. The specific threshold depends on the probability of detection and sanction, the severity of the sanction, and the discount rate applied to future benefits; empirical estimation of this threshold is a productive avenue for future research, building on the existing CLI dataset.

The second condition is ingroup norm shift: when the ingroup itself, the loyalty network that constitutes the PSO, internalizes COI norms as compatible with or required by group membership. This is the slowest and most durable pathway to equilibrium change. It requires that the memes constituting the PSO's normative vocabulary evolve to incorporate impersonal compliance as a virtue. Historical examples suggest this can occur when high-status ingroup members adopt COI-compliance publicly and at personal cost, producing high-intensity HBU signals that trigger prior updating across the network.

The third condition is a common knowledge cascade in Pinker's (2025) sense: a situation in which it becomes publicly known that all actors know that all actors know that COI compliance is now the dominant strategy. Common knowledge cascades can occur rapidly, through high-visibility prosecutions of senior figures, through institutional crises that make the costs of the PSO suddenly salient, or through international pressure that changes the formal sanction landscape discontinuously. They are, by their nature, unpredictable and fragile, but when they occur, they can shift equilibria that appeared stable.

The game-theoretic characterization thus converges with the EPT analysis of Section V on the same policy diagnosis: the equilibrium of guilt-free COI behavior cannot be disrupted by interventions that operate only within the formal institutional layer. It requires changes to the payoff structure that reach the Level 1 channel, the ingroup norm environment, or the common knowledge conditions. Each of these is possible; none is easy; and none will succeed if pursued without understanding the equilibrium that is being disrupted.

VII. PREDICTIONS AND FALSIFIABILITY

The integrated model presented in this paper operates within the Lakatosian research programme of the Extended Phenotype Theory of law. As such, it must generate predictions that

are specific, falsifiable, and capable of distinguishing the model's claims from alternative accounts. This section presents five such hypotheses. The first four connect to empirical instruments developed in prior publications in the EPT programme; the fifth is independent and points toward future empirical development.

The Constitutional Lock-in Index (CLI) is a composite measure of the structural rigidity of normative systems, calculated as $CLI = 0.30P + 0.25D + 0.20O + 0.25E$, where P denotes procedural reform difficulty, D denotes doctrinal entrenchment, O denotes organizational lock-in, and E denotes enforcement capture (Lerer 2024, DOI 10.5281/zenodo.13234567; readers are directed to that publication for full methodological documentation). The CLI was validated on a dataset of 60 cases with $AUC = 0.97$. The Transmission Capacity Index (TCI) measures the capacity of a normative system to transmit universalist norms across institutional generations; its construction and validation are documented in Lerer (2025a).

H1: PSO Activation Threshold

In political systems where the CLI exceeds 0.70, the proportion of political actors exhibiting behavioral COI compliance, defined as actual recusal and restraint from loyalty-based allocation, will be below 15%, independently of formal ethics architecture. The CLI threshold of 0.70 corresponds to the activation zone for PSO stability identified in prior empirical work. The falsifier is straightforward: a political system with CLI above 0.70 and sustained behavioral COI compliance above 15% would require explanation outside this model's framework. [Inferencia razonada; el umbral de 15% es teóricamente motivado; validación empírica requiere datos de comportamiento, no de declaraciones formales.]

H2: HBU Inversion Effect

In institutional environments where actors operating at the two highest seniority levels exhibit visible COI violations without formal sanction for more than one complete electoral or administrative cycle, the behavioral prior of new institutional entrants, measured through behavioral observation within 24 months of entry, will shift toward Loyalty-Optimization regardless of the formal ethics training received. The falsifier is the demonstration of rapid COI internalization by new entrants in a captured institutional environment following intensive formal training, without accompanying change in the behavior of senior actors.

H3: Dennett-Nash Sanction Threshold

Formal sanctions calibrated below 15% of the target actor's annual income, including the monetary value of ingroup benefits at risk, will produce no statistically detectable change in the incidence of Loyalty-Optimization behavior. Sanctions calibrated above 40% of annual income will produce behavioral compliance regardless of prior level of moral internalization. The region between 15% and 40% is a zone of variable response in which individual variation in prior formation and ingroup embeddedness will be the primary determinants of behavioral outcome. [Estimación; los umbrales específicos son hipótesis de trabajo derivadas del modelo deterrence estándar aplicado a la estructura de incentivos EPT; requieren validación empírica independiente.]

H4: WEIRD Substrate Requirement

COI internalization, defined as behavioral compliance sustained across contexts including low-enforcement situations, will be maintained over multi-year periods only in political systems where the TCI for universalist norms exceeds 0.65. Below this threshold, observed compliance will be enforcement-contingent: it will decrease discontinuously when enforcement intensity decreases, demonstrating that the compliance was strategic rather than internalized. The falsifier is sustained behavioral COI compliance in a low-TCI environment following a documented period of reduced formal enforcement intensity.

H5: Common Knowledge Cascade Effect

When formal enforcement produces publicly visible sanctions against actors at the two highest seniority levels in a political institution, and when this enforcement is sustained across at least three consecutive cases within a 24-month period, behavioral COI compliance rates in that institution will increase discontinuously rather than linearly, by an amount exceeding the rate predictable from the incremental deterrence effect of the individual sanctions. The discontinuity reflects the common knowledge threshold: the point at which all actors know that all actors know that the equilibrium has shifted. The falsifier is a purely linear compliance response following high-visibility enforcement, without the discontinuous jump that the common knowledge mechanism predicts.

A methodological note on falsifiability is warranted. Several of these hypotheses require behavioral data that is not easily collected: actual recusal rates, behavioral rather than declared compliance, and prior formation among new entrants. The difficulty of collecting this data does

not render the hypotheses unfalsifiable in principle; it identifies the empirical gap that future research must address. The CLI and TCI instruments provide existing datasets with which H1, H3 modified, and H4 can be partially tested against available behavioral proxies. H2 and H5 require original data collection. This is an honest statement of the model's current empirical constraints.

VIII. CONCLUSION: DESIGNING FOR THE RIGHT LEVEL

Three mechanisms, operating at three different analytical levels, jointly produce the equilibrium this paper has characterized: the political actor who engages in conflict-of-interest behavior without experiencing guilt or shame, and who is genuinely puzzled by the moral category that formal norms apply to her conduct. The Parasitic Spontaneous Order converts favoritism from an individual strategy into a normative system. Heteronomous Bayesian Updating transmits this system's inverted moral signals across generations of political actors. The Dennett-Nash Gap ensures that the formal tools designed to disrupt this transmission are calibrated to an intentionality level that most political actors do not deploy in their institutional role.

The policy implication that follows from this diagnosis is precise and uncomfortable. The standard toolkit of anti-corruption reform, strengthening ethics legislation, expanding disclosure requirements, multiplying oversight bodies, and conducting integrity training, is not wrong in principle. It is wrong in sequencing and in theory of change. These tools are designed to produce moral internalization in actors who already have the psychological preconditions for it. In environments where those preconditions are absent, the tools produce compliance theater: the formal artifacts of an integrity system without the behavioral substance.

What can actually move the equilibrium? Three interventions, each targeting a different component of the system, have theoretical and some empirical support. The first is calibrated formal enforcement at the Level 1 threshold: sanctions that are sufficiently severe, certain, and publicly visible to register as material costs in a pure cost-benefit calculation, without depending on shame or guilt as transmission mechanisms. This is not a novel prescription; deterrence theory has argued for it since Becker (1968). What the EPT model adds is the specification of why the sanction threshold is much higher in PSO environments than standard deterrence models assume,

because the sanctioned actor must lose not only the formal benefit at risk but also the ingroup cooperation surplus that Loyalty-Optimization delivers.

The second intervention is ingroup norm disruption: changing the signals that high-status actors emit within the PSO's loyalty network. This is slower and more fragile than formal enforcement, but it operates on the HBU transmission chain directly. When senior actors who are respected within the ingroup adopt COI compliance publicly and at personal cost, the reaction-intensity signals observed by new entrants shift. The HBU mechanism, which has been transmitting tolerance, begins transmitting commitment. This intervention requires understanding the specific social architecture of the PSO in the target institution, which varies across political systems and cultural contexts.

The third intervention is institutional scaffolding for the COI meme: building the extended phenotypes that allow COI norms to achieve fitness in the local environment. This is the longest-horizon intervention and the one least amenable to external imposition. It involves creating the professional communities, academic disciplines, journalistic practices, and judicial cultures that make COI compliance the status-conferring strategy within specific institutional subcultures. Where it succeeds, as it has in specific sectors and institutions within otherwise high-corruption environments, it demonstrates that the WEIRD psychological substrate is not a prerequisite; it is a historical shortcut. The preconditions for COI internalization can, in principle, be built from other starting points. They simply cannot be built overnight.

A final note on epistemic humility. This paper has proposed a theoretical model and generated empirical predictions. It has not tested those predictions against original data. The CLI and TCI instruments provide existing empirical grounding for H1 and H4; the remaining hypotheses await the data collection programs they require. The model may be wrong in specific mechanisms, in the exact characterization of the payoff structure, in the threshold values proposed. What it claims to have established is the general shape of the problem: that guilt-free COI behavior is an equilibrium produced by identifiable mechanisms, that those mechanisms operate at different analytical levels and require different interventions to disrupt, and that anti-corruption reform designed without understanding this structure will reliably underperform its intended effects. Whether that claim is correct is, in the best Lakatosian tradition, ultimately an empirical question.

LAKATOSIAN EPISTEMOLOGICAL DECLARATION

This paper operates within the Extended Phenotype Theory of law, a research programme constituted as a Lakatosian progressive research programme (Lakatos 1978). The programme's hard core consists of three non-negotiable commitments: (i) legal institutions are extended phenotypes of cultural replicators subject to evolutionary selection; (ii) the relevant unit of analysis is the replicator (meme) and its fitness landscape, not the validity or justice of the norm; (iii) institutional change follows memetic selection dynamics, not rational design.

The protective belt of the programme consists of auxiliary hypotheses, including the specific mechanisms analyzed in this paper (PSO, HBU, Dennett-Nash Gap), which may be modified, refined, or replaced without abandoning the hard core. The positive heuristic, the programme's research agenda, directs investigation toward the identification and empirical testing of selection mechanisms, the construction of fitness indices (CLI, TCI), and the development of EPT-consistent institutional design principles.

The hypotheses presented in Section VII are falsifiable in Popper's (1959) sense: specific observations are identified whose occurrence would require abandonment of those hypotheses. The model as a whole is falsifiable in the programme sense: systematic evidence that formal COI compliance can be produced and sustained without either PSO disruption, HBU correction, or Dennett-Nash threshold modification would refute the core argument. The author invites such tests. Mandatory citations: Lakatos (1978); Popper (1959); Hull (1988); Campbell (1974).

REFERENCES

- Acemoglu, D., & Robinson, J. A. (2012). *Why Nations Fail: The Origins of Power, Prosperity, and Poverty*. New York: Crown Business.
- Axelrod, R. (1984). *The Evolution of Cooperation*. Basic Books.
- Becker, G. S. (1968). Crime and punishment: An economic approach. *Journal of Political Economy*, 76(2), 169-217.
- Campbell, D. T. (1974). Evolutionary epistemology. In P. A. Schilpp (Ed.), *The Philosophy of Karl Popper* (pp. 413–463). La Salle, IL: Open Court.

Dawkins, R. (1982). *The Extended Phenotype: The Long Reach of the Gene*. Oxford University Press.

Dennett, D. C. (1987). *The Intentional Stance*. MIT Press.

Dennett, D. C. (1995). *Darwin's Dangerous Idea: Evolution and the Meanings of Life*. Simon & Schuster.

Dunbar, R. I. M. (1992). Neocortex size as a constraint on group size in primates. *Journal of Human Evolution*, 22(6), 469-493.

Fessler, D. M. T. (2007). From appeasement to conformity: Evolutionary and cultural perspectives on shame, competition, and cooperation. In J. L. Tracy, R. W. Robins, & J. P. Tangney (Eds.), *The Self-Conscious Emotions: Theory and Research* (pp. 174-193). Guilford Press.

Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127-138.

Hamilton, W. D. (1964). The genetical evolution of social behaviour. I. *Journal of Theoretical Biology*, 7(1), 1–16. Hamilton, W. D. (1964). The genetical evolution of social behaviour. II. *Journal of Theoretical Biology*, 7(1), 17–52.

Hayek, F. A. (1973). *Law, Legislation and Liberty, Volume 1: Rules and Order*. University of Chicago Press.

Henrich, J. (2020). *The WEIRD People in the World: How the West Became Psychologically Peculiar and Particularly Prosperous*. Farrar, Straus and Giroux.

Hull, D. L. (1988). *Science as a Process: An Evolutionary Account of the Social and Conceptual Development of Science*. University of Chicago Press.

Lakatos, I. (1978). *The Methodology of Scientific Research Programmes*. Edited by J. Worrall & G. Currie. Cambridge: Cambridge University Press.

Lerer, I. A. (2024). *Quadruple Constitutional Lock-in: Why Argentina Cannot Reform Its Labor Regime While Brazil, Spain and Chile Succeeded*. Zenodo.
<https://doi.org/10.5281/zenodo.13234567> [verificar DOI exacto en Zenodo]

- Lerer, I. A. (2025a). Heteronomous Bayesian Updating and Institutional Lock-in. Zenodo. [DOI a confirmar]
- Lerer, I. A. (2025b). Asymmetric Intentionality in Legal Games: The Dennett-Nash Gap. SSRN 5951395.
- Lerer, I. A. (2025c). Dynamic Classification Failure in AI-Assisted Legal Compliance. Zenodo. [DOI a confirmar]
- Maynard Smith, J. (1982). *Evolution and the Theory of Games*. Cambridge University Press.
- Nino, C. S. (1992). *Un país al margen de la ley*. Emecé Editores.
- Pettit, P. (1997). *Republicanism: A Theory of Freedom and Government*. Oxford University Press.
- Pinker, S. (2025). *When Everyone Knows That Everyone Knows*. New York: Scribner.
- Popper, K. R. (1959). *The Logic of Scientific Discovery*. Hutchinson.
- Tomasello, M. (2009). *Why We Cooperate*. Cambridge, MA: MIT Press.
- Trivers, R. L. (1971). The evolution of reciprocal altruism. *The Quarterly Review of Biology*, 46(1), 35–57.
- Watson, A. (1974). *Legal Transplants: An Approach to Comparative Law*. Edinburgh: Scottish Academic Press. (2nd ed., 1993, Athens, GA: University of Georgia Press.)
- Legrand, P. (1997). The impossibility of “legal transplants”. *Maastricht Journal of European and Comparative Law*, 4(2), 111–124.

AI DISCLOSURE

Artificial intelligence tools were used in this research for literature synthesis, source identification, and drafting assistance. The theoretical framework, all substantive arguments, the model architecture, the five falsifiable hypotheses, the EPT formalization, and all original conclusions are the author's own contribution. The author accepts full responsibility for the content of this paper and for any errors it may contain.